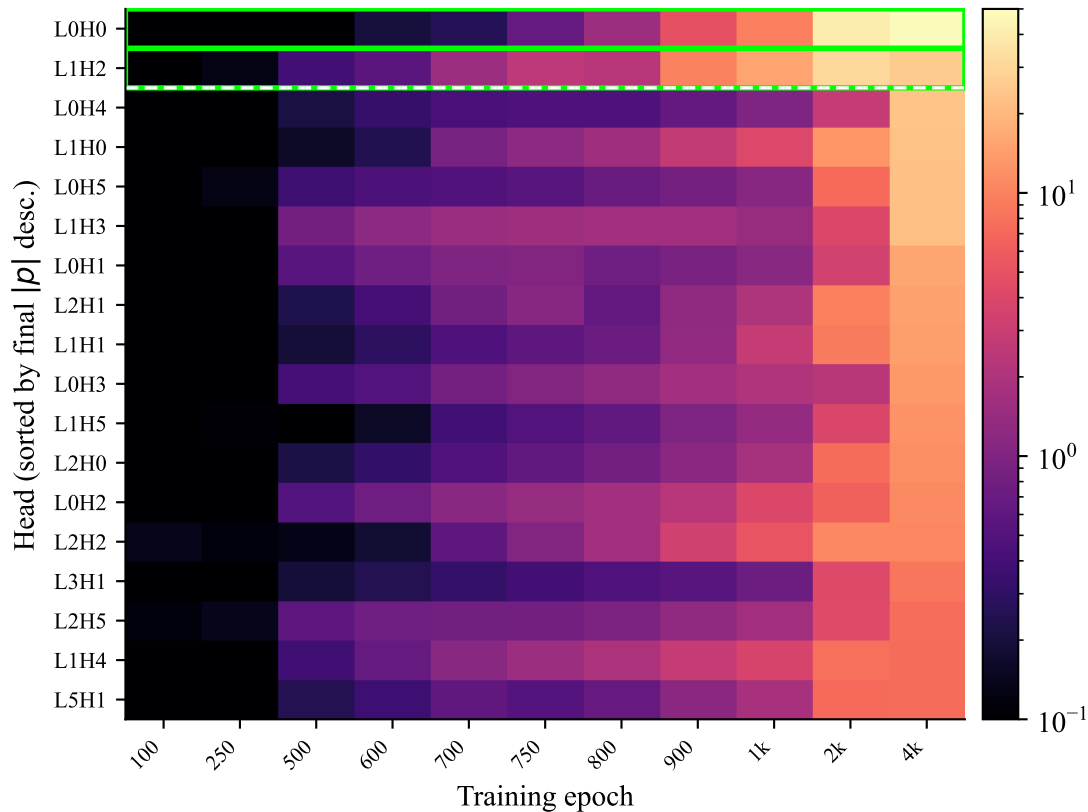


# Spatial-relation alignment across training: $L_0H_0$ and $L_1H_2$ emerge sharply

(a) Per-head emergence: projection magnitude  $|p|$



(b) Phase-transition emergence at epochs 750–1000

